Unreal Engine Cheat Sheet

UE 5.8 · hotfix 5.8.4 · checked 5 November 2026 · Irurueta Docs, Graphics & Games

Versions and licence

Item	Status
Current	**5.8** (17 Jun 2026), hotfix **5.8.4** (20 Oct 2026)
Previous	5.7 (Nov 2025)
UE6	Epic targets Early Access at the end of 2027
Royalty	Free to US$1M lifetime per product, then **5%**; Epic Games Store revenue exempt
Launch Everywhere	**3.5%**, launch on the Store first or at the same time
Seat licence	US$1,850 per user per year, non-game companies above US$1M a year

Project layout and build tools

```
Bookshelf.uproject   version, modules
Source/Bookshelf/    Build.cs, .h, .cpp
Source/*.Target.cs   build target
Content/             .uasset, .umap
Config/              Default*.ini
Plugins/             more modules
Binaries Intermediate Saved
DerivedDataCache     generated, no Git
```

UHT	generates reflection code from macros
UBT	compiles modules and targets
UAT	runs jobs such as BuildCookRun
Live Coding	Ctrl+Alt+F11, patches a running editor

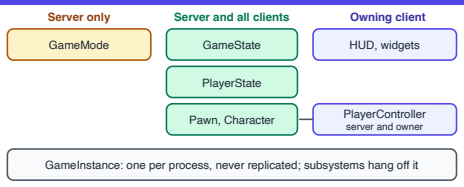
Specifiers

Macro	Common specifiers
`UCLASS`	`Blueprintable`, `BlueprintType`, `Abstract`, `Config=Game`
`UPROPERTY`	`EditAnywhere`, `EditDefaultsOnly`, `VisibleAnywhere`, `BlueprintReadWrite`, `BlueprintReadOnly`, `Category`, `Replicated`, `ReplicatedUsing`, `Transient`
`UFUNCTION`	`BlueprintCallable`, `BlueprintPure`, `Server`, `Client`, `NetMulticast`, `Reliable`, `Exec`

Garbage collection rules

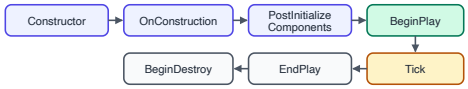
- A `UObject` lives while reachable from a root through `UPROPERTY` references
- **Rule:** mark every `UObject*` member `UPROPERTY()`; write `TObjectPtr<T>`
- `TWeakObjectPtr` observes, `TSoftObjectPtr` loads later
- `TSharedPtr` is for non-`UObject` data only
- Create with `NewObject` and `SpawnActor`, never `new`
- `IsValid(p)` also catches pending kill

Gameplay framework: who has what



The controller possesses the Pawn. The server decides; clients ask through RPCs.

Actor lifecycle order



Do not touch other actors in the constructor. Clear timers and delegates in `EndPlay`.

Blueprint and C++ exposure

Keyword	Means
`BlueprintCallable`	C++ body, BP calls it
`BlueprintPure`	as above, no exec pin
`BlueprintImplementableEvent`	C++ calls, BP implements
`BlueprintNativeEvent`	BP may override `Foo_Implementation`
`BlueprintAssignable`	dispatcher BP can bind

Blueprint communication

Need	Use
Know the one target	Direct reference
Need its class members	Cast (creates a hard reference)
Many classes, same message	Interface
One sender, many listeners	Event dispatcher

Enhanced Input

Raw key → **Modifiers** (dead zone, negate, swizzle) → **Triggers** (pressed, hold, tap) → **Input Action**. A **Mapping Context** holds the bindings and is added per local player with a priority.

```
Subsystem->AddMappingContext(Context, 0);
Input->BindAction(MoveAction,
  ETriggerEvent::Triggered, this,
  &ABookshelfCharacter::Move);
void Move(const FInputActionValue& V) {
  FVector2D A = V.Get<FVector2D>();
}
```

Rendering toggles

Feature	Where, status in 5.8
Nanite	per mesh; `r.Nanite 0` to compare
Lumen	Project Settings, Dynamic GI and Reflection Method; default
Lumen Lite	medium quality GI, beta
MegaLights	production-ready; many shadowed lights
VSM	`r.Shadow.Virtual.Enable`; default
Mobile	multi-pass deferred by default; `r.Mobile.ShadingPath`
Scalability	`sg.ShadowQuality 0` to 4

Material essentials

- Inputs: Base Color, Metallic, Roughness, Normal, Emissive, Ambient Occlusion
- **Parameter** + **Material Instance**: change values without recompiling
- Dynamic instance at run time: `SetVectorParameterValue`
- Material Functions share graphs; Substrate is the layered model; Substrate NPR is experimental

Collision

Each body has an **Object Type** and a response to each channel: Ignore, Overlap or Block. Default channels: WorldStatic, WorldDynamic, Pawn, PhysicsBody, Vehicle, Destructible; traces use Visibility and Camera.

Event	Needs
`OnComponentHit`	Block on both, Simulation Generates Hit Events
`OnComponentBeginOverlap`	Overlap, Generate Overlap Events on both

```
GetWorld()->LineTraceSingleByChannel(
  Hit, Start, End, ECC_Visibility);
```

Replication

- Roles: `ROLE_Authority`, `ROLE_AutonomousProxy`, `ROLE_SimulatedProxy`
- `bReplicates = true`; register in `GetLifetimeReplicatedProps` with `DOREPLIFETIME`
- `ReplicatedUsing=OnRep_X`: RepNotify runs on clients

RPC	Runs on
`Server`	server, from the owning client
`Client`	the owning client
`NetMulticast`	server and all relevant clients

AI stack

Move	NavMesh, `AAIController`
Sense	AI Perception, not PawnSensing
Decide	StateTree, or behaviour tree + Blackboard
Query	EQS; Smart Objects to use the world
Crowds	Mass Entity

Console commands

`stat unit`	Frame, Game, Draw, GPU times
`stat gpu`	GPU passes
`stat fps`	frame rate
`r.ScreenPercentage 70`	GPU bound?
`trace.start`	Unreal Insights capture
`showdebug`	HUD debug text

Packaging and Android

```
RunUAT BuildCookRun -project=...
  -platform=Win64 -clientconfig=Shipping
  -build -cook -stage -pak -package
  -archive
```

Android Studio	Koala 2024.1.2 Patch 1
NDK, JDK	r27c, OpenJDK 21.0.3
SDK	35 recommended, 34 minimum; install 26+
Google Play	target **API 36** from 31 Aug 2026; upload an AAB

Set Target SDK Version yourself: Epic's pages give 34 or 35.

Old to new

Old	New
Matinee	Sequencer
Cascade	Niagara
PhysX	Chaos
Legacy input	Enhanced Input
World Composition	World Partition
Hot Reload	Live Coding
PawnSensing	AI Perception